

Course syllabus

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(PLEASE consider the details below as preliminary until the course begins.)

(If this page is the only page you can see, and you are currently participating in the course, you have not been properly registered. Then please send a mail to Ashkan Panahi (see below for the mail address), and we will register you!)

Course-PM / Syllabus

DAT410 / DIT728 DAT410 / DIT728 Design of AI systems lp3 VT26 (7.5 hp)

Course is offered by the department of Computer Science and Engineering

Introduction

The purpose of the course is to explain how some different well-known AI-systems work, provide insight in how such systems are built, and practice to develop such systems. The course takes a broad perspective and includes related areas such as data science, algorithms and optimization as appropriate.

The main structure of the course is given by the weekly modules:

1. **AI problem solving** [Ashkan]
2. **Recommendation systems** [Bastiaan]
3. **AI tools** [Filip]
4. **AI for Medicine and Health** [Farzaneh]
5. **Computer Vision and Remote Sensing** [Shirin]
6. **Games and planning** [Jingjing]
7. **Dialogue systems and question answering** [Ashkan]

8. Your own mini-project [Ashkan]

There may additionally be single lectures that are not connected to a particular module.

The learning objectives are found in the [official course plan](#). The course plans for the different course codes are essentially the same.

For detailed information about the modules see the Modules page.

Contact details

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Contact us via email or Canvas messages. If you respond to the comments in Canvas we might not see it.

Student Representatives

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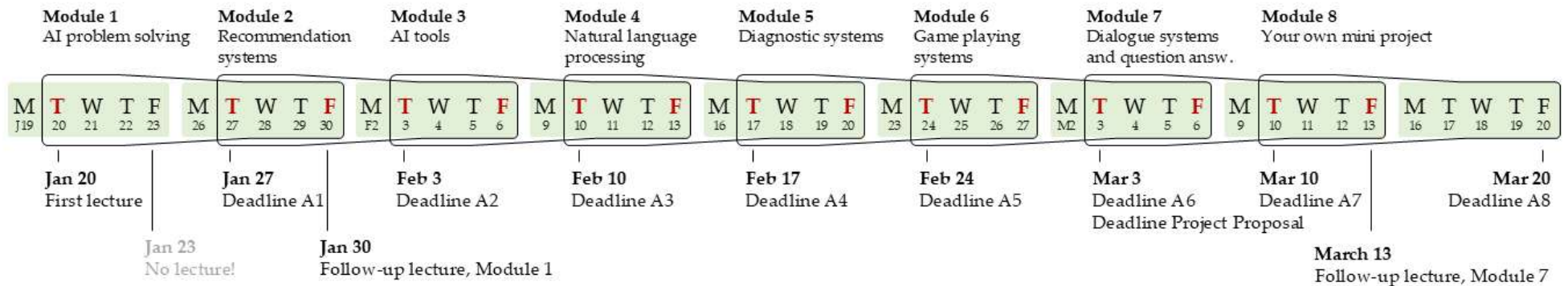
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Schedule

The course will be in the physical format. Some material from the previous years might be provided online.

The schedule may vary slightly from week to week, see the [TimeEdit schedule](#) and specific information for each module. Below you can find an overview of the course:



Examination and grading

The course is examined continuously through the module submissions. You pass the course by passing every module in the course.

All assignments must be submitted as a report saved in PDF format. The report should include coherent text and well-formulated answers to each question, clearly organized with each answer corresponding to its respective question. Append your code to the end of the report. While we will visually inspect

A report satisfying what we ask for in the assignment will achieve a passing grade, to receive a higher grade the following grading criteria will be assessed:

- Accuracy and coherence in summary of reading material
- Justification and quality of problem solution
- Clarity in description of solution/implementation
- Quality in critical assessment/evaluation of solution
- Discussion of results

All text and codes that you submit must be your original work. If you have to include text from any other source, it must be well cited. In particular, it is not allowed to include a text or code which is generated by language models such as GPT.

The first two modules of the course are simply pass/not pass. The remaining modules are graded based on a qualitative assessment on the scale Sufficient, Good and Very Good related to the Chalmers and GU grades in the following way:

- **Sufficient** (3/G/40 in Canvas)
- **Good** (4/G/60 in Canvas)
- **Very Good** (5/VG/80 in Canvas)

The Canvas numbers are a way to numerically encode the qualitative grade in the Canvas grade field, and have no direct relationship to any proportion of correctly answered questions. For a normal Sufficient grade we would write 50, but the interval 40-59 is possible - the lowest possible passing level for a module is 40. As other examples, 75 indicates an upper-level "Good" and 80 indicates a borderline grade between Good and Very Good. The Canvas code 8 is used to indicate that the submission has not yet passed.

The final grade is based on a weighted sum of the graded modules. The final grade limits in Canvas are 40 (Sufficient/3/G), 60 (Good/4/G) and 80 (Very Good/5/VG). If you are close to a grade boundary, you can discuss with the main teacher at the end of the course, and you will be given an opportunity to improve (there is no point in doing this as the course proceeds, only at the end). At GU, only final grades G and VG are available. **Please note that before all assignments have passed, the final Canvas score may be misleading.**

If you should not complete the course in time, and need to come back next year, it is in your best interest to keep copies of your solutions to enable future assessment.

Grading will be done by the teaching assistants (see below). If you have questions regarding the assignments, please contact the respective TAs that are responsible for that assignment directly.

		Grading								Consultation	
Module	Topic	Filip	Hang	Shirin	Christopher	Yaochen	Nils	Jingjing	Yingjue	Thurs (v-1) 10:00– 12:00	Mon (v) 10:00– 12:00
1	AI problem solving	X				X				Yaochen	Filip

2	Recommendation systems	X				X				Yaochen	Filip
3	AI tools	X	X			X				Filip	Hang
4	Diagnostic systems		X					X	X	Jingjing	Hang
5	CV and Remote Sensing		X	X			X			Nils	Shirin
6	Game playing systems				X		X	X		Christopher	Nils
7	Dialogue systems and question answering				X		X		X	Nils	Christopher
8.0	Mini project proposal	X	X	X	X	X	X	X	X		
8	Mini project	X	X	X	X	X	X	X	X		

Consultation sessions

Assignment consultation sessions will be held on **Zoom on Thursdays 10:00–12:00 and Mondays 10:00–12:00**. Please see the schedule above which TA is responsible and use the corresponding Zoom link (see below) to contact them. It may take some time for you to be admitted from the waiting room to the meeting, so please hold. Please **only join the meeting during the above-mentioned hours**. Also **double check which TA is responsible** for which consultation session! The same applies when asking questions via mail.

Filip: <https://chalmers.zoom.us/j/7353696169>

Shirin: <https://chalmers.zoom.us/j/68261115177>

Christopher: <https://chalmers.zoom.us/j/67226789929>

Nils: <https://gu-se.zoom.us/j/7446007263>

Jingjing: <https://chalmers.zoom.us/j/68148423212>

Hang: <https://chalmers.zoom.us/j/2051704086>

Yaochen: <https://chalmers.zoom.us/j/5509224936>

Yingjue:

No password is needed to enter the Zoom meetings, make sure you are logged in with your Chalmers/GU account, however.

Mini-projects

There will be consultation sessions only for the mini-project proposals on Thursday and Monday. See above who's Zoom meeting you should join:

TA	Group Number	Note
Shirin		
Filip		
Hang		
Yaochen		
Nils		
Christopher		
Yingjue		
Jingjing		

Course literature

There is no compulsory course literature. Reading instructions will be provided in connection to the modules.

Changes made since the last occasion

All assignments are now required to be in pdf format. You cannot submit a copy of a notebook. All text must be in a proper, readable format. You can only copy the codes to the pdf file (e.g. from your notebook).

Course summary:

Date	Details	Due
Tue, 27 Jan 2026	☐ Assignment 1: AI Problem Solving	due by 23:59
Tue, 3 Feb 2026	☐ Assignment 2: Recommendation systems	due by 23:59
Tue, 10 Feb 2026	☐ Assignment 3: AI Tools (CORRECT)	due by 23:59
	☐ Assignment 3: AI Tools (DO NOT SUBMIT!)	due by 23:59
Tue, 17 Feb 2026	☐ Assignment 4: Diagnostic Systems	due by 23:59
Tue, 24 Feb 2026	☐ Assignment 5: Semantic Segmentation with Deep Learning	due by 23:59
Tue, 3 Mar 2026	☐ Assignment 6: Game playing systems	due by 23:59
	☐ Assignment 8.0: Your mini-project proposal	due by 23:59
Tue, 10 Mar 2026	☐ Assignment 7: Dialogue systems and question answering	due by 23:59
Fri, 20 Mar 2026	☐ Assignment 8: Your own mini-project	due by 23:59